

Null tests of the LOS closure-bound estimator: real turbulence, ODT, and a diagnosed initial-condition artifact

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Summary. The LOS estimator (previous note: forward map, realizability cone, per- κ_2 linear programs) has now been exercised on two kinds of real line data: (i) an ensemble of lines through DNS of forced isotropic turbulence (JHTDB), and (ii) ensembles of nominal-HIT ODT realizations run on DLR CARO with the strain machinery active and $A_{ij} = 0$ — your false-positive test. Outcome in one sentence: *the estimator produces no spurious strain signal on either data source; it did, however, catch a real $\sim 1.5\%$ component-anisotropy bias in the ODT initial condition, which we attributed by a reversal experiment and eliminated by a code fix* — the eddy mechanism itself passes the null test cleanly.

1 Pipeline

Identical for both data sources. From an ensemble of velocity lines $\{u_i(y)\}$: cross line-spectra $\phi_{mn}(\kappa_2)$ (all six components) with jackknife errors; the on-line A1 diagnostics (T1: $\phi_{11} = \phi_{33}$; T2: vanishing cross spectra); the classical isotropic longitudinal–transverse derivative relation (T3); and the LP bounds $[\Pi^-, \Pi^+]$ at three constraint levels (raw; polarization cap $\gamma = 0$; $\gamma = 0$ plus log-Lipschitz slope cap $\lambda = 4$), with the measured standard errors entering as inequality bands (T4). Component energies give b_{ij} (T5). ODT lines use the interior 80% segment with a power-normalized Hann window (WALL boundaries); JHTDB lines are periodic.

2 Real-turbulence null: JHTDB isotropic

128 lines (1024 points each) through one snapshot of `isotropic1024coarse`. Results: T1 splitting exceeds 2σ in 7% of wavenumber bins (5% expected by chance); T2 coherences ~ 0.01 ; T3 holds to $\approx 4\%$ (quadrature-limited); T4: the bounds bracket the Crow value $\Pi_{ij}^{(r)} = \frac{4}{5}k_t S_{ij}$ at every constraint level with widths consistent with the synthetic isotropic calibration. The component energies give the **statistical noise floor**: with 128 lines from one snapshot, b_{ij} scatters at ± 0.06 (dominated by the few independent large-scale structures per snapshot). Strain-generated anisotropy in the regime of interest is $b \sim 0.2\text{--}0.4$, so the floor is comfortable, and it shrinks as $1/\sqrt{N}$ with snapshots.

3 ODT null: nominal HIT on CARO, and what it caught

Case: the manuscript’s homogeneous-strain configuration with $A_{ij} = 0$, dilatation off, eddies on ($\nu = 10^{-4}$, $C = 7$, $Z = 450$, band-limited isotropic IC, $k_t(0) = 1$); 1024 independent realizations per variant (≈ 5 minutes wall time each ensemble on one CARO node).

What passed immediately. The estimator brackets the Crow value at all constraint levels and all times; T1/T2 spectral nulls are clean; the $t = 0$ control is isotropic ($b_{33} = +0.004 \pm 0.003$).

What the pipeline caught. From $t = 0.5$ onward the component energies showed a small but statistically solid bias: $b_{33} = +0.0148 \pm 0.0046$ (3.2σ), $b_{11} < 0$, persistent in sign at

every dump time (figure, left panel). Two facts pointed away from the eddy dynamics and toward the initial condition: the pattern was *monotone in the order* (u, v, w) , and that is exactly the order in which the IC whitening was applied — an ordered Cholesky factor of the empirical 3×3 covariance, which leaves u a pure random-phase field, orthogonalizes v against u , and corrects w against both.

Attribution by reversal. Rebuilding the initializer with the whitening order permuted to (w, v, u) — nothing else changed — flips the pattern: the b_{33} signal collapses to $+0.002 \pm 0.004$ and the over-energized slot moves to u (figure, middle panel). The artifact follows the ordering, not the physics.

Fix and confirmation. The whitening was replaced by the symmetric inverse square root $C^{-1/2}$ (Jacobi eigen-decomposition of the 3×3 covariance), which treats the components identically by construction. With the fixed initializer all three components are within $\approx 1\sigma$ of zero at every time ($\max |b_{ii}| = 0.005$; figure, right panel), and every spectral null is clean (T1 exceedances: 0%).

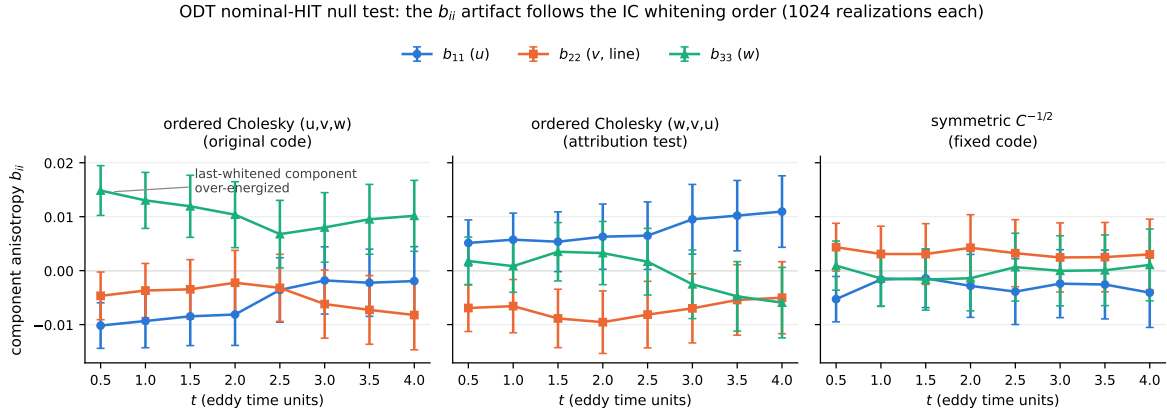


Figure 1: Component anisotropy $b_{ii}(t)$ in nominal-HIT ODT, 1024 realizations per panel, jackknife error bars. Left: original ordered Cholesky IC whitening — the last-whitened component (w) is over-energized at the 1–1.5% level and the first (u) under-energized. Middle: reversing the whitening order flips the pattern, attributing the bias to the IC, not the eddy dynamics. Right: symmetric $C^{-1/2}$ whitening (the code fix) — all components consistent with isotropy.

4 Implications

1. **ODT’s eddy mechanism generates no false strain signal** in nominal HIT at the $\sim 0.5\%$ level our 1024-realization ensembles resolve — the direct answer to the false-positive question, now with the IC artifact removed rather than argued away.
2. **The estimator pipeline is validated end-to-end on real data**, including its error model: the same code that will process strained-box DNS lines and strained ODT lines has now run on both data types, bracketed the known answers, and demonstrated enough sensitivity to find (and localize) a 1% systematic.
3. **The artifact episode is itself the strongest sensitivity statement:** a component bias twenty times smaller than the strain signals of interest was detected, causally attributed by a designed intervention, and eliminated. The percent level is exactly where the DNS-comparison campaign will operate.

4. **Bookkeeping for the manuscript:** the published runs used the ordered-Cholesky IC. The $\sim 1\%$ bias is far below the strain-generated anisotropy studied there and does not threaten those conclusions, but the fixed initializer should be used for all future runs, and the change noted.

Next: the strained-box DNS (plane strain, $Sk/\varepsilon \in \{0.8, 4, 16\}$, $e \leq 2$, CARO) supplies the exact $\Pi_{ij}^{(r)}$, the true axisymmetric scalars and their azimuthal residues, and the LOS bound widths under real distortion — the quantitative test of the “weakly constrains” question that these null tests have now cleared the runway for.